

Conversational Search Cost Optimization Recommendations

Content Generation program, guest facing conversational search

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Guest facing conversational search currently runs on Gemini 3.1 Flash-Lite through the TIP LiteLLM proxy at a modeled run rate of about 537,000 dollars per month. About 75 percent of that is recoverable at equal quality, in a prescriptive order: cache the fixed prefix, prune and compact context, then swap the default model to gemma4:31b on Vertex, with a small hard tail that escalates only on a verified quality miss. Every recommendation below is backed by a measured result from our routing and cost studies, and every token saving is shown with its calculation. The appendix carries the full derivations, assumptions, and evidence tables.

1. Executive summary

The cost driver is structural, not the model: a 15,000 token system prefix and the retrieved context are re-sent on every turn, and history grows, so input tokens scale roughly quadratically with conversation length. Input is about 98 percent of tokens and about 89 percent of cost. That is why caching the fixed prefix is the single largest safe lever, before any model change. The recommended sequence and its expected savings:

The one new lesson

The hard tail should be an escalate on failure cascade, not an up front difficulty predictor. Our agentic routing study showed that predicting which inputs are hard does not work once tasks get complex; running the cheap model and escalating only on a verified miss does. Conversational search is easier to verify than an agentic task, so the existing grounding check doubles as the escalation trigger.

Lever	What it does	Expected saving	Quality risk
1. Prefix caching	Bill the re-sent system and static RAG prefix at the cached rate	about 35 percent	None
2. Pruning and compaction	Trim the system prompt and RAG, compact history	to about 57 percent	Low, gated
3. Default model swap	gemma4:31b on Vertex, measured on par with Flash-Lite	to about 75 percent	Gated, non inferior
4. Hard tail escalation	Escalate only verified misses to Flash-Lite or Pro	protects quality	Reduces risk

2. The cost problem

Each turn re-sends the full instruction and tool block plus retrieved context, and also carries the growing conversation history. For a turn t , the input is the fixed prefix plus the history accumulated from prior turns. With a 15,000 token system prompt, 4,000 tokens of RAG, a 40 token query, and 400 tokens of output per turn, the per turn input grows by 440 tokens each turn (one query plus one output of carried history).

$$I(t) = \text{system} + \text{rag} + (\text{query} + \text{output}) (t - 1) + \text{query}$$

$$I(t) = 19,000 + 440 (t - 1) + 40 \text{ tokens}$$

2.1 Per turn token breakdown

Turn	System	RAG	Carried history	Query	Input tokens
1	15,000	4,000	0	40	19,040
2	15,000	4,000	440	40	19,480
3	15,000	4,000	880	40	19,920
4	15,000	4,000	1,320	40	20,360
Total	60,000	16,000	2,640	160	78,800

3. Baseline forecast

Volume is modeled from 270 million Bonvoy members at 3 percent monthly adoption and 3 conversations per active searcher per month. These inputs are explicit and auditable, and should be calibrated against Adobe event206 to event208 volume and TIP token telemetry before external use. At Flash-Lite list pricing of 0.25 dollars input and 1.50 dollars output per million tokens, the baseline is about 537,000 dollars per month.

$$N_{\text{conv}} = 270,000,000 \times 0.03 \times 3.0 = 24,300,000 \text{ conversations/month}$$

$$C_{\text{conv}} = \frac{78,800 \times \$0.25 + 1,600 \times \$1.50}{10^6} = \frac{\$19,700 + \$2,400}{10^6} = \$0.0221$$

$$C_{\text{month}} = \$0.0221 \times 24,300,000 = \$537,030$$

3.1 Baseline at a glance

Quantity	Value
Conversations per month	24,300,000
Input tokens per conversation	78,800
Output tokens per conversation	1,600
Cost per conversation	0.0221 dollars
Input tokens per month	1.915 trillion
Output tokens per month	38.9 billion
Monthly run rate	537,030 dollars

4. Lessons learned that shape these recommendations

These recommendations are not generic best practice; each rests on a measured result from the LiteForge routing paper, the cross provider cost and quality study, and the agentic routing study (all in the appendix and the docs folder). The table maps each finding to the decision it drives here.

Measured finding	Decision for conversational search
gemma4:31b scored 0.866 vs Flash-Lite 0.859 head to head (paired diff +0.007, 95 percent CI -0.017 to +0.031): statistically on par.	Swap the default to gemma4:31b on Vertex behind a non inferiority gate.
Cross provider frontier: gemma4:31b reaches 0.865 (96 percent of Sonnet) and on published serverless rates costs 0.78 dollars per million, undercutting Flash-Lite.	The swap is a quality preserving cost cut, corroborated by a second independent measurement.
Agentic study: gemma4:31b as the cheap tier beat a 397B model at 11 times lower cost.	A strong cheap base model is the dominant lever; lead with model selection.
Predictive difficulty routing scored at or below random on agentic tasks (APGR -0.14 and -0.44).	Do not build an up front difficulty router for the hard tail.
Escalate on failure beat always using the strongest model (oracle cascade 0.727 vs 0.628).	Make the hard tail a cascade that escalates only on a verified miss.
Verification quality scales with the judge and is the bottleneck (best recall about 0.60).	Gate escalation on the existing grounding check, which is a strong cheap verifier for QA.

5. Recommendations

Five recommendations, in execution order. Each states the finding it rests on, the decision, the calculation, and the expected saving. Savings are cumulative in the stated order and are shown for the expected scenario unless noted.

5.1 R1. Cache the fixed prefix

Finding and decision: the system prompt and the static share of RAG are identical across turns, so they can be billed at the cached rate on a hit. This is the largest lever with zero quality risk. Ship it first. The cacheable prefix per conversation is the fixed block re-sent on each of the four turns.

Cache hit rate	Monthly cost	Saved
50 percent	418,112 dollars	22.1 percent
60 percent	394,328 dollars	26.6 percent
70 percent	370,545 dollars	31.0 percent
80 percent	346,761 dollars	35.4 percent
90 percent	322,977 dollars	39.9 percent
95 percent	311,086 dollars	42.1 percent

$$P_{\text{cache}} = 4 \times (15,000 + 0.6 \times 4,000) = 4 \times 17,400 = 69,600 \text{ of } 78,800 \text{ input tokens}$$

5.2 R2. Prune context and compact history

Finding and decision: not all of the prefix is load bearing. Trimming the system prompt and tightening retrieval shrink the re-sent block, and compacting history caps the quadratic growth term.

Applied after caching, the expected scenario (40 percent system trim, 35 percent RAG trim, 60 percent history compaction) reaches about 57 percent cumulative. This is gated by the quality check in the next section.

after prune (expected) = \$232,114 $\Rightarrow$ 56.8% saved cumulatively

5.3 R3. Swap the default model to gemma4:31b on Vertex

Finding: gemma4:31b is on par with Flash-Lite on quality by three independent measurements (head to head 0.866 vs 0.859; cross provider frontier 0.865 at 96 percent of Sonnet; and as the agentic cheap tier it beat a 397B model). Decision: make it the default for nearly all traffic on managed Vertex, in GCP, per token. At the input heavy mix it is about 42 percent cheaper at equal quality, taking the expected scenario from about 57 percent to about 75 percent.

$$r_{\text{save}} = \frac{0.2749 - 0.159}{0.2749} = 0.42 \Rightarrow 42\% \text{ cheaper at equal quality}$$

5.4 R4. Escalate the hard tail on verified failure

Finding: in the agentic study, predicting which inputs need the strong model failed (APGR at or below random), but running the cheap model and escalating only on a verified miss beat even always using the strongest model. The cheap realistic trigger was self consistency; a model judge helped only when strong and calibrated. Decision: keep gemma4:31b as the default, and escalate a turn to Flash-Lite or Gemini Pro only when the grounding and faithfulness check fails, with self consistency as the free fallback trigger. Conversational search is easier to verify than an agentic booking, so a cheap verifier suffices here. The tail adds back only the escalated share.

$$C_{\text{with tail}} = (1 - r) C_{\text{cheap}} + r C_{\text{strong}}, \quad r = \text{verified miss rate}$$

5.5 R5. Build the levers once in TIP

Decision: caching, pruning, the default model policy, the escalation cascade, and spend metrics belong in the TIP SDK as a shared platform capability, not re-implemented per application. Conversational search is the first and highest volume tenant; the same capability then serves Content Generation consumers and other agents, and emits spend to the control plane that governs the routing and guardrail configuration.

6. Savings curve

Three operating points let leadership choose against effort and risk. Conservative applies caching and pruning only. Expected adds the gemma4:31b default swap. Aggressive pushes caching, pruning, and routing harder. Dollar figures are monthly run rate from the model.

Stage	Conservative	Expected	Aggressive
Baseline	537,030	537,030	537,030
After caching	457,751	346,761	265,897
After pruning	376,456	232,114	144,707
After model swap	376,456	138,055	75,160
After hard tail	376,456	135,294	72,154
Total saved	30 percent	75 percent	87 percent

7. Quality gate and rollout

No change ships unless it is strictly non inferior to the Flash-Lite baseline on real conversational search queries, scored by an LLM judge and a grounding and faithfulness check on Lakshmi Nim-magadda's eval set, with bootstrap confidence intervals. The gemma4:31b swap already clears non inferiority on public benchmarks (paired diff +0.007, CI -0.017 to +0.031). Recommended rollout is shadow, then a gated A and B test with automatic rollback on any KPI regression.

Calibration before external use

These are model outputs, not measured spend. Calibrate adoption and turns from Adobe event206 to event208, the token mix from TIP and Dynatrace telemetry, the in tenant Flash-Lite and Vertex prices and the cached token discount, and reconcile the baseline against the Egor and Kenny cost alignment numbers.

8. Appendix

The receipts: every assumption, derivation, scenario calculation, and the measured evidence behind each recommendation. Numbers regenerate from the scripts named in section K.

8.1 A. Assumptions and sources

Parameter	Value	Source to calibrate against
Bonvoy members	270,000,000	Loyalty reporting
Monthly adoption	3 percent	Adobe event206 to event208 (validate)
Conversations per user per month	3.0	Adobe analytics (validate)
Turns per conversation	4.0	TIP conversation telemetry
System prompt tokens	15,000	Confirmed, instruction and tool block
RAG tokens per turn	4,000	Retrieval config (60 percent static)
Query tokens per turn	40	Telemetry
Output tokens per turn	400	Telemetry
Flash-Lite price in and out	0.25 and 1.50 per 1M	Jira GENAI-1412, WEB-194505; validate in tenant
Cached read multiplier	0.25	Vertex context caching (confirm support)

8.2 B. Baseline derivation in full

Input per turn is the fixed prefix plus carried history plus the query; the four turns sum to 78,800 input tokens, with 1,600 output tokens. Monthly volume is 24.3 million conversations.

$$I(1) + \dots + I(4) = 19,040 + 19,480 + 19,920 + 20,360 = 78,800$$

$$\text{tokens/month} = 78,800 \times 24,300,000 = 1.915 \times 10^{12} \text{ input}$$

$$C_{\text{month}} = \frac{78,800(0.25) + 1,600(1.50)}{10^6} \times 24,300,000 = \$537,030$$

8.3 C. Caching derivation

On a cache hit the cacheable prefix is billed at the cached multiplier m of the input price. With cacheable tokens c , hit rate h , input I and output O , the per conversation cost is the formula below. The 50 to 95 percent hit sweep in recommendation R1 follows directly, from 22.1 percent to 42.1 percent saved.

$$C = \frac{(I - ch)p_{in} + chp_{in}m + Op_{out}}{10^6}, \quad c = 69,600, \quad m = 0.25$$

8.4 D. Pruning and compaction derivation

System trim and RAG trim reduce the fixed prefix; history compaction reduces the quadratic carried term. The expected scenario uses 40 percent system trim, 35 percent RAG trim, and 60 percent compaction, reaching 56.8 percent cumulative with caching.

$$I_{\text{pruned}}(t) = \text{sys}(1 - s) + \text{rag}(1 - r) + (q + o)(t - 1)(1 - k) + q$$

$$s = 0.40, \quad r = 0.35, \quad k = 0.60 \Rightarrow \$232,114/\text{month}$$

8.5 E. Model swap price derivation

Blended price is the input output mix weighted rate. At the conversational search mix gemma4:31b on Vertex is 42 percent cheaper than Flash-Lite at on par quality, which moves the expected scenario from 56.8 percent to 74.3 percent.

$$p_{\text{blend}} = \frac{p_{in}T_{in} + p_{out}T_{out}}{T_{in} + T_{out}} : 0.2749 \rightarrow 0.159 \Rightarrow 42\% \text{ lower}$$

8.6 F. Per scenario cumulative savings

Scenario	Cache	Plus prune	Plus model	Plus tail
Conservative	14.8 percent	29.9 percent	29.9 percent	29.9 percent
Expected	35.4 percent	56.8 percent	74.3 percent	74.8 percent
Aggressive	50.5 percent	73.1 percent	86.0 percent	86.6 percent

8.7 G. Provider cost and quality table (corrected)

From the cross provider study, n equals 290 public benchmark prompts. OSS prices are published serverless rates (Together, DeepInfra, Mistral) blended at this token mix, fetched 2026-06-27. These replace an earlier owned compute estimate that was throughput noise (it had a 14B model at 23 dollars and a 4B above a 120B).

Model	Accuracy	Percent of Sonnet	Dollars per 1M
Sonnet 4.6 (anchor)	0.896	100	11.27 measured
gemma4:31b	0.865	96	0.78 published
glm-5.2	0.861	96	4.08 published
Gemini 3.1 Flash-Lite	0.858	96	1.05 measured
deepseek-v4-pro	0.854	95	3.31 published
DeepSeek v4 Flash	0.850	95	0.26 measured
gpt-oss-120b	0.844	94	0.49 published
gemma3-27b	0.771	86	0.14 published
Mistral Small	0.741	83	0.09 measured

8.8 H. Routing and cascade evidence

Why the hard tail is a cascade, not a predictor. Predictive difficulty routing scored at or below random on agentic tasks; escalate on failure beat even always strong; and a verifier helps only when strong and calibrated. Conversational search is QA like, so a lightweight router is viable for an easy and hard split, but the robust mechanism is the grounding gated cascade.

Strategy or router	Metric	Result
Embedding head on RouterBench (QA)	APGR	0.31
router-bert on agentic	APGR	-0.14
MF port on agentic	APGR	-0.44
All strong (glm-5.2), gemma pool	quality	0.628
Oracle cascade, gemma pool	quality	0.753
Self consistency cascade, gemma pool	quality	0.613
Judge cascade (glm-5.2)	quality	0.614
Judge recall, gemma3:27b to glm-5.2	recall	0.07 to 0.60

8.9 I. Cost model formulas

The exact functions used. Per conversation tokens accumulate the re-sent prefix and the growing history; cost applies the cached multiplier to the hit fraction of the cacheable prefix.

$$\text{input} = \sum_{t=1}^T [\text{sys} + \text{rag} + (\text{q} + \text{o})(t - 1) + \text{q}]$$

$$\text{cost} = \frac{(\text{input} - \text{ch}) p_{\text{in}} + \text{ch} p_{\text{in}} m + \text{output} p_{\text{out}}}{10^6}$$

8.10 J. Glossary and terminology

Term	Meaning
Content Generation	The program. ConEng is the TIP Context Engine component built for it; the two are not interchangeable.
APGR	Average performance gain recovered: 0 is random routing, 1 is the oracle.
Cascade	Run the cheap model first, escalate to a stronger model only on a verified miss.
Self consistency	Sample the cheap model a few times and escalate when the samples disagree.
Blended price	Input output mix weighted dollars per million tokens at this workload.
Non inferiority	Strictly not worse than baseline within a confidence interval (the do no harm gate).

8.11 K. References and reproducibility

- Forecast model: scripts/forecast/conv_search_forecast.py, results/forecast.json.
- Cross provider study: docs/provider-cost-quality-study.md, results/provider_frontier.json.
- Agentic routing study: docs/agentic-routing-study.md (cascade, verifier, gemma findings).
- Routing paper: docs/paper/liteforge-routing.tex (Studies 1 to 4).
- Forecast narrative: docs/conv-search-token-forecast.tex.
- Provider prices fetched 2026-06-27: together.ai, deepinfra.com, mistral.ai.
- Setup tickets: Jira GENAI-1412 and WEB-194505 (Flash-Lite via the TIP proxy).

Bottom line

Lead with model selection and structural savings: cache the prefix, prune and compact, set gemma4:31b on Vertex as the default, and escalate only verified misses. Expected recovery is about 75 percent of a 537,000 dollar per month run rate at equal quality, every number shown in the appendix.

Appendix and evidence

The appendix that follows carries the full assumptions, derivations, per scenario calculations, the corrected provider cost and quality table, and the routing and cascade evidence with confidence intervals.